Strict negativity and non-vanishing of Chen–Larson hypergeometric coefficients

A computer-assisted and formally verified proof of Propositions 5.1 and 5.2 (the latter corrected)

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Abstract

Chen and Larson reduce a geometric vanishing problem for strata of holomorphic abelian differentials to the (non-)vanishing of a coefficient in a quotient of hypergeometric generating series. We treat both series their argument produces. For the residue classes $g \equiv 0, 2 \pmod 3$ covered by their Proposition 5.1 we prove the stronger statement that the relevant coefficient is always strictly negative: with

$$\mathsf{C}(t) = \sum_{r \geq 0} \frac{(6r)!}{(3r)!(2r)!} \left(\frac{t}{72}\right)^r,$$

if $g \geq 2$, $g \equiv 0, 2 \pmod 3$, $a = \lfloor g/3 \rfloor + 1$, and $\mu = (m_1, \ldots, m_n)$ is a positive partition of $2g-2$, then

$$[t^a] \frac{\prod_{i=1}^n \mathsf{C}(t/(m_i+1))}{\mathsf{C}(t)^{2g-2+n}} < 0.$$

For the complementary class $g \equiv 1 \pmod 3$, governed by their Proposition 5.2, we first *correct* the printed relation: the constant $\kappa_0 = 2g-2$ of Ionel's relation was inadvertently replaced by 1, which changes the coefficient. We then prove that the corrected coefficient is non-vanishing for every positive partition, and strictly negative for $a \geq 14$. Together the two results settle the coefficient (non-)vanishing in all three residue classes, hence in every genus, beyond the authors' computer check ($g \leq 30$).

The Proposition 5.1 proof removes the partition by a coefficientwise majorization; a sign-lock estimate with an explicit rational final margin then shows that all convolution indices above $0.9a$ contribute nonpositively, and the remaining positive part is controlled by a direct saddle method based on finite exponential prefixes. Small ranges are discharged by exact enumeration and verified rational or dyadic interval certificates; the large range is closed by symbolic ratio and reserve bounds. The corrected Proposition 5.2 reduces, through a one-line identity, to two sign inputs already controlled by the Proposition 5.1 analysis, together with an exhaustive finite check. The complete argument — both propositions and the correction — has been formalized in Lean 4 and Mathlib, the public theorems depending only on the standard axioms and the two compiler-trust axioms of `native_decide`. We also describe the role of long-horizon generative-AI assistance in the development, including an AI audit that uncovered the Proposition 5.2 error in the original paper, and the precise trust boundary of the computational certificates.

Contents

1 Introduction

1.1 Geometric context

Chen and Larson study tautological classes on strata of differentials. In Section 5 of their work [1], they pull back a relation among κ-classes on $\mathcal{M}_g$ to a stratum associated with a partition $\mu = (m_1, \dots, m_n)$ of $2g-2$. For the residue classes $g \equiv 0, 2 \pmod 3$, their Proposition 5.1 reduces the desired vanishing to a single coefficient of the series

$$\frac{\prod_{i=1}^{n} \mathsf{C}(t/(m_i+1))}{\mathsf{C}(t)^{2g-2+n}}, \qquad \mathsf{C}(t) = \sum_{r \geq 0} \frac{(6r)!}{(3r)!(2r)!} \left(\frac{t}{72}\right)^r. \tag{1}$$

For a positive partition $\mu = (m_1, \dots, m_n)$ of $2g-2$ — a tuple of integers $m_i \geq 1$ summing to $2g-2$ — let $\mathcal{P}(\mu)$ denote the projectivized stratum of holomorphic abelian differentials whose zero orders are $m_1, \dots, m_n$, and let η denote its tautological line-bundle class, in the convention of Chen–Larson. Their Conjecture 1.4 predicts a nontrivial tautological relation in the first degree not excluded by their independence theorem, namely $a = \lfloor g/3 \rfloor + 1$; for $g \equiv 0, 2 \pmod 3$, Proposition 5.1 reduces the geometric vanishing $\eta^a = 0$ to the nonvanishing of the single coefficient in (1). The complementary class $g \equiv 1 \pmod 3$ is governed by their Proposition 5.2, which uses a different, derivative-corrected series; we treat it — in corrected form — in Section 9. The geometric statement requires only that the relevant coefficient be nonzero; for Proposition 5.1 our main result gives a uniform sign.

Theorem 1.1 (Main theorem)**.** *Let $g \geq 2$ satisfy $g \equiv 0$ or $2 \pmod 3$, and set*

$$a = \lfloor g/3 \rfloor + 1.$$

For every positive partition $\mu = (m_1, \dots, m_n)$ of $2g-2$,

$$[t^a] \frac{\prod_{i=1}^{n} \mathsf{C}(t/(m_i+1))}{\mathsf{C}(t)^{2g-2+n}} < 0. \tag{2}$$

In particular, the coefficient in Chen–Larson Proposition 5.1 is nonzero.

The strict negativity is stronger than the nonvanishing needed geometrically; we attach no geometric meaning to the sign itself.

Corollary 1.2. *In the setting of Chen–Larson Proposition 5.1, the class η^a vanishes for every positive holomorphic partition when $g \equiv 0, 2 \pmod 3$.*

Proof. Combine Theorem 1.1 with Chen–Larson Proposition 5.1. □

For the remaining residue class we prove a non-vanishing statement for the *corrected* Proposition 5.2 series; the correction itself is explained in Section 9.

Theorem 1.3 (Corrected Proposition 5.2)**.** *Let $g \equiv 1 \pmod 3$, write $g = 3a - 2$ so that $M := 2g - 2 = 6a - 6$, and let $\mu = (m_1, \dots, m_n)$ be a positive partition of M. With*

$$B_\mu(t) = \frac{\prod_{i=1}^n \mathsf{C}(t/(m_i+1))}{\mathsf{C}(t)^N}, \quad N = \sum_i (m_i+1), \qquad K_\mu(t) = \sum_{i=1}^n m_i\, \Phi(t/(m_i+1)), \quad \Phi(t) = 2t + 12t^2 \frac{\mathsf{C}'(t)}{\mathsf{C}(t)},$$

the corrected Proposition 5.2 coefficient $[t^a]\, B_\mu(t)\big(M - K_\mu(t)\big)$ is nonzero for every $a \geq 2$, and strictly negative for $a \geq 14$. The series printed by Chen and Larson is the same expression with the leading constant M replaced by 1.

Corollary 1.4. *Combining Theorems 1.1 and 1.3 with the geometric reductions of Chen–Larson Propositions 5.1 and 5.2 (the latter corrected as in Section 9), the class $\eta^{\lfloor g/3 \rfloor + 1}$ vanishes for every positive holomorphic partition, in* every *genus $g \geq 2$.*

1.2 Outline

After the geometric reduction, the statement is no longer one of algebraic geometry: it is a uniform claim about a single coefficient of a quotient of power series. Two features make it hard. The quotient depends on an arbitrary partition $\mu = (m_1, \dots, m_n)$ of $2g - 2$, and the same strict sign is required for every genus in the two residue classes under consideration and for every such partition. The proof meets these in turn: first it removes the partition, leaving an expression in two integer parameters; then it proves the sign for that expression uniformly — in the large-a argument, without expanding the coefficients themselves (in the finite range they are computed exactly). This subsection describes the argument in words; every object named here is defined precisely in the sections that follow.

Removing the partition. The one structural fact we need about C is that its logarithm has nonnegative coefficients: writing

$$\log \mathsf{C}(t) = \sum_{r \geq 1} c_r t^r,$$

one has $c_r > 0$ (proved in Section 3). Set $q_i = m_i + 1 \geq 2$ and $N = \sum_i q_i = 2g - 2 + n$; the coefficient of interest is $[t^a] \prod_i \mathsf{C}(t/q_i)/\mathsf{C}(t)^N$. Rescaling t inside the logarithm writes each factor of the numerator as an exponential, and the product as a single exponential,

$$\mathsf{C}(t/q_i) = \exp\Big(\sum_{r \geq 1} c_r\, q_i^{-r}\, t^r\Big), \qquad \prod_{i=1}^n \mathsf{C}(t/q_i) = \exp\Big(\sum_{r \geq 1} c_r\, \sigma_r\, t^r\Big), \quad \sigma_r := \sum_{i=1}^n q_i^{-r}.$$

The coefficients of an exponential $\exp(\sum_r a_r t^r)$ are polynomials in the a_r with nonnegative coefficients; since here $a_r = c_r \sigma_r \geq 0$, each coefficient of the product is a nondecreasing function of every

σ_r. Now $q_i \geq 2$ gives $\sigma_r \leq n\,2^{-r}$, and $n \leq N/2$, so $\sigma_r \leq \frac{N}{2}\,2^{-r}$; replacing σ_r by this larger value turns the product into exactly $\mathsf{C}(t/2)^{N/2}$. Hence the numerator is dominated coefficientwise,

$$[t^r]\prod_i \mathsf{C}(t/q_i) \;\leq\; [t^r]\,\mathsf{C}(t/2)^{N/2} \qquad (r \geq 0),$$

by an upper bound that no longer remembers the partition. Substituting it for the numerator and dropping the convolution terms that are automatically nonpositive produces a partition-free upper bound $U_a(N)$ for the target coefficient — a *majorant* — whose explicit form is recorded in (8). It then remains to prove

$$U_a(N) < 0 \qquad \text{for the two integers } a, N \text{ subject to } 6a - 7 \leq N \leq 12a - 8.$$

A negative main term against small positive corrections. It is convenient to rescale the coefficients of the two relevant series. Let $X_r(N)$ be the normalized coefficient of t^r in the denominator $\mathsf{C}(t)^{-N}$, and $Y_r(N)$ the normalized coefficient of t^r in $\mathsf{C}(t/2)^{N/2}$ (the precise normalizations are fixed in Section 2). In these terms the majorant takes the shape

$$\frac{U_a(N)}{Nc_a} = \underbrace{X_a(N)}_{\text{one negative term}} + \text{(positive correction terms)}.$$

The proof is a contest between the two sides: we bound $X_a(N)$ above by a *negative* number with an explicit margin, and bound every positive correction below that margin. The corrections come from the convolution and are indexed by splittings $a = k + j$ with $1 \leq k < a$.

Most splittings contribute nothing. We prove a *sign lock*: once $k > 0.9a$, the coefficient $X_k(N)$ is already negative throughout the whole range of N, so such terms only help an upper bound and may be dropped. This is the technical heart of the negative side, and it leaves only the splittings with $1 \leq k \leq \lfloor 0.9a \rfloor$ to be controlled.

Bounding coefficients without computing them. For the surviving corrections we never expand the coefficients as numbers — that would be hopeless for large a. Instead we use one elementary principle. If a series is the exponential of a series with nonnegative coefficients, its coefficient of t^m is bounded, for any positive rescaling of t, by a finite truncation of an ordinary exponential,

$$P_m(x) = \sum_{r=0}^{m} \frac{x^r}{r!}.$$

Choosing the rescaling well makes this bound sharp; this is the classical saddle-point idea, but carried out so that every quantity stays rational[1] and no transcendental constant enters. Two finite exponentials multiply in the same spirit,

$$P_m(x)\,P_n(y) \leq P_{m+n}(x+y) \qquad (x, y \geq 0),$$

[1]This emphasis on staying rational reflects the formal proof more than the mathematics. The direct-saddle estimates and the final numerical certificate inequalities are formulated over $\mathbb{Q}$; quantities that an informal asymptotic argument might estimate with Stirling's formula or the real exponential are bounded by finite rational inequalities. One short auxiliary bridge in the sign-lock argument of Section 5 does use formal estimates for the real logarithm and exponential to control the factorial product Π_s, after which the argument returns to rational inequalities; no estimate involving π is used. The slight loss in the resulting constants is absorbed by the large slack in the final budget.

which is the binomial theorem applied after enlarging a rectangle of indices to a triangle. This last inequality is more important than it looks: an early formalization attempt bounded the two factors $\mathsf{C}(t)^{-N}$ and $\mathsf{C}(t/2)^{N/2}$ separately and multiplied, which discarded a large power of 2 and produced a false target (Section 11). The proof that works keeps the product together and combines the finite exponentials only at the very end.

Finite checking and the infinite tail. The argument is split deliberately between explicit finite verification and uniform inequalities. For $a \le 8$ every partition is enumerated and the exact rational coefficient is checked. For $9 \le a \le 60$ the majorant $U_a(N)$ is checked exactly over the rectangle, and for $61 \le a \le 400$ the same check is run in verified interval arithmetic — a small calculator for intervals of rationals whose soundness is itself proved (so that rounding can never certify a false inequality). For all $a \ge 401$ the sign lock supplies the negative margin and the saddle bounds reduce the positive side to a tiny budget: the strip $401 \le a \le 2000$ is verified directly, and the strip $2001 \le a < 3000$ together with the genuinely infinite range $a \ge 3000$ are closed by showing that consecutive terms shrink by a fixed rational ratio, so each tail is at most its first term times a geometric series. The positive budget never exceeds 10^{-8}, while the negative margin is strictly larger; hence $U_a(N) < 0$, and the Chen–Larson coefficient is negative.

1.3 Formal verification and scope

The theorem is formalized over $\mathbb{Q}$ in Lean 4 and Mathlib. The public interface names the hypergeometric series, proves its coefficient formula, characterizes the quotient generating series by the identity in (1), and exposes strict negativity and non-vanishing. The exact public theorem is reproduced in Appendix B.

The finitely many explicit computations — enumerations, exact rational checks, and interval certificates — are discharged by Lean's `native_decide`, which verifies a finite claim by compiling it to a program and running it. This makes the compiler part of what one trusts for those steps, in exchange for being able to check millions of cases. The trade is recorded openly: Lean's `#print axioms` command lists exactly which axioms a given theorem rests on, and for our final theorem that list is the standard logical axioms of Mathlib together with two named native-evaluation axioms and nothing else. Section 10 gives the precise trust boundary; the rest of the proof is checked by Lean's small logical kernel in the usual way.

2 The partition-free majorant

Let $\mu = (m_1, \dots, m_n)$ be a positive partition of $2g-2$ and put

$$q_i = m_i + 1 \ge 2, \qquad N = \sum_{i=1}^{n} q_i = 2g - 2 + n. \tag{3}$$

Set

$$b_a(\mu) = [t^a] \frac{\prod_i \mathsf{C}(t/q_i)}{\mathsf{C}(t)^N}. \tag{4}$$

For $g \equiv 0, 2 \pmod 3$ and $a = \lfloor g/3 \rfloor + 1$, one has

$$g = 3a - 3 \quad \text{or} \quad g = 3a - 1.$$

Since $1 \le n \le 2g-2$, every relevant value of N lies in the enlarged rectangle

$$6a - 7 \le N \le 12a - 8. \tag{5}$$

The enlargement is convenient because it treats both residue classes at once.

Define

$$\begin{aligned} P_r(\mu) &= [t^r] \prod_i \mathsf{C}(t/q_i), \\ B_r(N) &= [t^r]\mathsf{C}(t)^{-N}, \\ Q_r(N) &= [t^r]\mathsf{C}(t/2)^{N/2}. \end{aligned} \tag{6}$$

Here rational powers are interpreted through the exponential of the formal logarithm. Since the logarithmic coefficients c_r are nonnegative, coefficients of

$$\exp\left(\alpha \sum_{r\ge 1} c_r 2^{-r} t^r\right)$$

are polynomials in α with nonnegative coefficients. Moreover, $q_i \ge 2$ and $n \le N/2$. It follows that

$$0 \le P_r(\mu) \le Q_r(N). \tag{7}$$

Expanding (4), dropping nonpositive convolution terms, and using (7) gives the partition-free majorant

$$b_a(\mu) \le U_a(N) := B_a(N) + Q_a(N) + \sum_{\substack{1\le k<a \\ B_k(N)>0}} B_k(N) Q_{a-k}(N). \tag{8}$$

Thus Theorem 1.1 follows once we prove

$$U_a(N) < 0 \tag{9}$$

throughout (5), except for the finitely many values where we instead verify $b_a(\mu) < 0$ directly.

2.1 Normalized form

For $r \ge 1$ define

$$X_r(N) = \frac{B_r(N)}{N c_r}, \qquad Y_r(N) = \frac{2^r Q_r(N)}{(N/2) c_r}, \tag{10}$$

and set

$$R_{k,a} = \frac{c_k c_{a-k}}{c_a}. \tag{11}$$

The standard exponential-coefficient recurrence gives

$$X_1 = -1, \qquad X_r = -1 - \frac{N}{r} \sum_{k=1}^{r-1} k R_{k,r} X_{r-k}, \tag{12}$$

$$Y_1 = 1, \qquad Y_r = 1 + \frac{N}{2r} \sum_{k=1}^{r-1} k R_{k,r} Y_{r-k}. \tag{13}$$

A direct calculation transforms (8) into

$$\frac{U_a(N)}{N c_a} = X_a(N) + 2^{-a-1} Y_a(N) + \frac{N}{2} \sum_{\substack{1\le k<a \\ X_k(N)>0}} R_{k,a} 2^{-(a-k)} X_k(N) Y_{a-k}(N). \tag{14}$$

The proof therefore consists of a uniform negative lower margin for $-X_a$ and an upper budget for the remaining two positive contributions.

3 Logarithmic coefficients

The hypergeometric series C satisfies a Riccati differential equation. Equating coefficients in the logarithmic derivative yields

$$c_1 = \frac{5}{6}, \qquad c_r = 6(r-1)c_{r-1} + \frac{6}{r}\sum_{i=1}^{r-2} i(r-1-i)c_i c_{r-1-i} \quad (r \geq 2). \tag{15}$$

In particular, $c_r > 0$. Write

$$c_r = 6^r (r-1)! d_r. \tag{16}$$

Then

$$d_r = d_{r-1} + \frac{1}{r}\sum_{i=1}^{r-2} \frac{d_i d_{r-1-i}}{\binom{r-1}{i}}. \tag{17}$$

The sequence d_r is nondecreasing and begins with $d_1 = d_2 = 5/36$.

The formal proof uses rational bounds throughout.

Lemma 3.1 (Uniform coefficient bounds)**.** *For every $r \geq 1$,*

$$\frac{5}{36} \leq d_r \leq \frac{4}{25}. \tag{18}$$

Consequently,

$$\frac{5}{36}6^r(r-1)! \leq c_r \leq \frac{4}{25}6^r(r-1)!. \tag{19}$$

Proof. The lower bound follows immediately from (17). For the upper bound, let $A_r = [t^r]\mathsf{C}(t)$ and $F_r = A_r/(6^r(r-1)!)$. One has $d_r \leq F_r$ and

$$\frac{F_{r+1}}{F_r} = 1 + \frac{5}{36r(r+1)}.$$

A rational telescoping argument bounds F_r by

$$\frac{F_3}{1-(5/36)(1/3-1/r)} \leq F_3\frac{108}{103} < \frac{4}{25}.$$

No estimate involving π is required. □

The reciprocal-binomial estimate

$$\sum_{i=1}^{p-1} \binom{p}{i}^{-1} \leq \frac{4}{p} \tag{20}$$

combines with (17) and (18) to give an increment bound.

Lemma 3.2 (Increment and ratio control)**.** *For $r \geq 2$,*

$$d_r - d_{r-1} \leq \frac{64/625}{r(r-1)}. \tag{21}$$

If $1 \leq s < m$, then

$$1 - \frac{2304}{3125}\frac{s}{m(m-s)} \leq \frac{d_{m-s}}{d_m} \leq 1. \tag{22}$$

Proof. The nonlinear summand in (17) is at most $(16/625)\binom{r-1}{i}^{-1}$. Summing with (20) gives (21). Telescoping from $m-s$ to m yields

$$d_m - d_{m-s} \le \frac{64}{625}\left(\frac{1}{m-s} - \frac{1}{m}\right).$$

Divide by $d_m \ge 5/36$. The upper ratio bound follows from monotonicity. □

A further consequence, used throughout the positive-part estimate, is

$$R_{k,a} \le \frac{576}{3125}\frac{1}{(a-1)\binom{a-2}{k-1}} \qquad (1 \le k < a). \tag{23}$$

Indeed, the d-factor contributes at most $(4/25)^2/(5/36) = 576/3125$, and the factorial quotient is exactly the reciprocal denominator in (23).

4 Finite verification through $a = 400$

The majorant is not strong enough for the smallest values of a, so the formal proof begins with direct finite verification.

4.1 Positive partitions for $a \le 8$

For $g \le 23$, equivalently $a \le 8$ in the relevant residue classes, every positive partition of $2g-2$ is generated and the exact rational coefficient $b_a(\mu)$ is evaluated. The largest enumeration is $p(44) = 75175$ partitions at $g = 23$. The executable theorem checks strict negativity for all of them. Independent cardinality statements verify selected partition counts, reducing the risk of an incomplete generator.

4.2 Exact majorant certificate for $9 \le a \le 60$

For $9 \le a \le 60$, the recurrence tables for c_r, $B_r(N)$, and $Q_r(N)$ are evaluated exactly over $\mathbb{Q}$. This verifies (9) at all 10764 pairs in the rectangle. As a regression anchor, the least-negative corner is recorded exactly as

$$\frac{U_9(100)}{100c_9} = -\frac{13391635371339739}{213818836571652096}. \tag{24}$$

4.3 Verified dyadic intervals for $61 \le a \le 400$

The remaining 470220 pairs are certified by a small executable interval kernel. A dyadic interval stores outward-rounded endpoints with a 192-bit mantissa. The exact rational recurrences are mirrored term by term, and separate soundness lemmas prove that the tables enclose the exact values. For the conditionally included term in (8), the checker uses the convex hull of the product interval and $\{0\}$; consequently it never needs to decide the sign of $B_k(N)$ from an interval approximation.

The interval kernel contains definitions only and imports no Mathlib code; its soundness theory is proved separately over $\mathbb{Q}$. Eight native-evaluation chunks cover the N-range. Their conjunction proves (9) for all $61 \le a \le 400$. The 192-bit precision matches the reference Arb calculation used during discovery, but Arb is not part of the trusted proof: Lean checks its own dyadic kernel and its rational enclosure semantics.

5 The effective sign lock

For $a \geq 401$, the leading term $X_a(N)$ is controlled analytically; more generally, the same estimate forces all sufficiently high convolution indices to be nonpositive. The argument rests on a single nonlinear estimate, which we isolate first.

5.1 The nonlinear residual envelope

Write the nonlinear part of the logarithm as

$$H(t) = \sum_{r\geq 2} c_r t^r, \qquad E_p^-(N) = [t^p]\exp\big(-N\,H(t)\big),$$

so that $\mathsf{C}(t)^{-N} = \exp(-Nc_1 t)\exp(-NH(t))$. Expanding this coefficient into factorial-composition blocks and bounding their total by the rational coefficient bounds of Section 3 and a rational form of Stirling's estimate yields the *normalized residual envelope*: for $m \geq 361$, $N \leq (40/3)m$, and $p \geq 2m/3$,

$$\left|\frac{E_p^-(N)}{-Nc_p} - 1\right| \leq \frac{66}{5m}. \tag{25}$$

In Lean this is `Eminus_normalized_residual_le_final` (`Prop51/Envelope.lean`), built on the H-power machinery of `Prop51/HPow.lean`.

5.2 The decomposition and the bound

Set $\zeta = 5N/(36m)$, so that $0 \leq \zeta \leq 50/27$ when $N \leq (40/3)m$. With $c_1 = 5/6$, expanding $B_m(N) = [t^m]\mathsf{C}(t)^{-N}$ by the s-fold contribution of the linear factor $\exp(-Nc_1 t)$ gives the exact finite identity

$$-X_m(N) = \sum_{s=0}^{m} \frac{(-Nc_1)^s}{s!}\left(-\frac{E_{m-s}^-(N)}{Nc_m}\right). \tag{26}$$

For $s < m$ the residual index $p = m - s$ is positive, and the summand rewrites, through the ratio c_{m-s}/c_m and (25), as

$$\frac{(-\zeta)^s}{s!}\,\Pi_s\,D_s\,(1+\varepsilon_{m-s}), \qquad \Pi_s = \prod_{i=1}^{s}\frac{1}{1-i/m}, \quad D_s = \frac{d_{m-s}}{d_m}, \quad \varepsilon_p = \frac{E_p^-(N)}{-Nc_p} - 1. \tag{27}$$

With $e_1(s) = s(s+1)/2$, writing $\Pi_s D_s(1+\varepsilon_{m-s}) = 1 + \big(e_1(s)-\zeta\big)/m + w_s$ and splitting the sum at $s = \lfloor m/3\rfloor$ turns (26) into three pieces:

$$\begin{aligned}-X_m(N) = {} & \underbrace{\sum_{0\leq s\leq\lfloor m/3\rfloor}\frac{(-\zeta)^s}{s!}\Big(1+\frac{e_1(s)-\zeta}{m}\Big)}_{\text{near base }\mathcal{B}_m} + \underbrace{\sum_{0\leq s\leq\lfloor m/3\rfloor}\frac{(-\zeta)^s}{s!}\,w_s}_{\text{near error}} \\ & + \underbrace{\sum_{\lfloor m/3\rfloor < s\leq m}\frac{(-Nc_1)^s}{s!}\left(-\frac{E_{m-s}^-(N)}{Nc_m}\right)}_{\text{far signed tail}}.\end{aligned} \tag{28}$$

In Lean the three pieces are named, respectively,

`signLockNearBase`, `signLockNearError`, `signLockFarSignedTail`,

all proved in `Prop51/SignLock.lean`. The residual envelope (25) applies on the near range $s \leq m/3$, where $p = m - s \geq 2m/3$.

The near base. The base is a *finite* sum, controlled without any appeal to an infinite exponential. Put

$$A_{12}(z) = \sum_{s=0}^{11} \frac{(-z)^s}{s!}, \qquad C_{12}(z) = \sum_{s=0}^{11} \frac{(-z)^s}{s!}(e_1(s) - z),$$

so the twelve-term prefix of $\mathcal{B}_m$ is $A_{12}(\zeta) + C_{12}(\zeta)/m$. Two Bernstein-basis certificates on $0 \le z \le 50/27$ give

$$A_{12}(z) \ge \lambda, \qquad A_{12}(z) + \frac{C_{12}(z)}{361} \ge \lambda\Big(1 - \frac{2}{361}\Big),$$

where

$$\lambda = \sum_{r=0}^{9} \frac{(-50/27)^r}{r!} = \frac{678107852315029}{4323713773987629} \tag{29}$$

(λ is defined by ten terms but used beneath the twelve-term certificate). A short monotonicity argument in $1/m$ extends the second inequality from $m = 361$ to every $m \ge 361$, and the base terms from $s = 12$ onward pair into nonnegative even–odd blocks; hence $\mathcal{B}_m \ge \lambda(1 - 2/m)$.

The near error and far tail. The other two pieces are bounded by seven independent rational budgets:

source	allowance in the numerator of m^{-2}
factorial-ratio residual	426
d-ratio drift	13
two-block recentering	184
higher two-block terms	234
three-or-more blocks	573
cross terms	784
far tails	1
total	2215

Every final allowance is a rational inequality. The factorial-ratio allowance uses a short real-logarithm and real-exponential bridge to control Π_s, after which the result is converted back into its rational budget. The near-error budgets are certified after rescaling ζ to $[0, 1]$ by nonnegative Bernstein coefficients, and the far signed tail (its first omitted $s = \lfloor m/3 \rfloor + 1$ block onward) is bounded by a crude rational saddle estimate for $E^-_{m-s}(N)$. Together they give a total error at most $2215/m^2$.

Theorem 5.1 (Sign lock)**.** *If $m \ge 361$, $N \ge 1$, and $N \le (40/3)m$, then*

$$X_m(N) \le -\left(\lambda\left(1 - \frac{2}{m}\right) - \frac{2215}{m^2}\right) < 0. \tag{30}$$

Proof. Combining $\mathcal{B}_m \ge \lambda(1 - 2/m)$ with the error bound $2215/m^2$ in (28) gives the displayed inequality once $\lambda(1-2/m) - 2215/m^2 > 0$. The latter is an exact rational check at $m = 361$ (where $m^2\lambda(1 - 2/m) > 2215$), and $m^2\lambda(1 - 2/m)$ is increasing in m, so it holds for every $m \ge 361$. □

Let

$$k_{\max}(a) = \lfloor 9a/10 \rfloor.$$

If $a \geq 401$ and $k > k_{\max}(a)$, then $k \geq 361$ and

$$N \leq 12a - 8 \leq \frac{40}{3}k.$$

Theorem 5.1 therefore implies $X_k(N) \leq 0$. Hence the positive sum in (14) can be restricted to

$$1 \leq k \leq \lfloor 9a/10 \rfloor . \tag{31}$$

For these retained indices, $j = a - k$ satisfies $j \geq a/10$.

6 Finite exponential prefixes and the direct saddle method

The formal proof does not estimate the relevant coefficients by invoking the real exponential directly. Instead, it works with finite prefixes

$$P_m(x) = \sum_{r=0}^{m} \frac{x^r}{r!}. \tag{32}$$

This keeps all intermediate statements rational.

6.1 Generic finite saddle lemmas

Let $L_r \geq 0$, $L_0 = 0$, and write

$$E_m(L) = [t^m] \exp\left(\sum_{r \geq 1} L_r t^r\right).$$

For $\rho \geq 0$, scaling gives

$$E_m(r \mapsto \rho^r L_r) = \rho^m E_m(L).$$

Expanding the exponential and bounding the coefficient of a power by the corresponding total gas yields the finite saddle inequality

$$\rho^m E_m(L) \leq P_m\left(\sum_{r=1}^{m} L_r \rho^r\right). \tag{33}$$

The other essential identity is the finite convolution bound.

Lemma 6.1 (Prefix convolution)**.** *For $x, y \geq 0$,*

$$P_m(x)P_n(y) \leq P_{m+n}(x+y). \tag{34}$$

Proof. Expand the left side over the rectangle $0 \leq p \leq m$, $0 \leq q \leq n$, enlarge to the triangle $p + q \leq m + n$, and group by $r = p + q$. The binomial identity gives

$$\sum_{p+q=r} \frac{x^p y^q}{p!q!} = \frac{(x+y)^r}{r!}.$$

All omitted summands are nonnegative. □

The same binomial argument proves

$$(1+d)^n \leq P_n(nd) \qquad (d \geq 0) \tag{35}$$

and the elementary absorption

$$\frac{x^r}{r!} \leq P_m(x) \qquad (0 \leq r \leq m,\ x \geq 0). \tag{36}$$

6.2 Factorial-gas estimates

The rational saddle radius is

$$\beta = \frac{34}{15}. \tag{37}$$

For $n \geq 40$ define

$$u_{n,r} = \beta^r \frac{(r-1)!}{n^r}.$$

Then

$$\frac{u_{n,r+1}}{u_{n,r}} = \frac{34r}{15n}. \tag{38}$$

Thus the sequence decreases and then increases (its consecutive ratio crosses 1 exactly once). Every term with $4 \leq r \leq n$ is bounded by $u_{n,4} + u_{n,n}$. The low and endpoint estimates formalized in the proof are

$$n(u_{n,2} + u_{n,3}) \leq \frac{1}{6}, \qquad n^2 u_{n,4} \leq \frac{1}{8}, \qquad n^2 u_{n,n} \leq \frac{1}{2}. \tag{39}$$

For the last inequality, set

$$W_n = n^2 u_{n,n} = \beta^n \frac{n!}{n^{n-1}}.$$

An exact check gives $W_{40} \leq 1/2$, and

$$\frac{W_{n+1}}{W_n} = \frac{\beta}{(1+1/n)^{n-1}} \leq 1 \qquad (n \geq 40).$$

Combining (38)–(39) gives

$$\sum_{r=2}^{n} \beta^r \frac{(r-1)!}{n^r} \leq \frac{1}{n}. \tag{40}$$

The proof has slack: after multiplication by n, its three budgets total $1/6 + 1/8 + 1/2 = 19/24$.

For the small branch, if $s \geq 32$ and $2 \leq k \leq s$, monotonicity of $(r-1)!/s^r$ after the first few terms gives

$$\sum_{r=2}^{k} \frac{(r-1)!}{s^r} \leq \frac{1}{s^2} + \frac{8}{s^3} \leq \frac{5}{4s^2}. \tag{41}$$

6.3 Coefficient saddle bounds

Let $B_k^+(N)$ denote the coefficient obtained from $\exp(N \sum_{r \geq 1} c_r t^r)$; in particular $B_k(N) \leq B_k^+(N)$. Take $s = \lceil \sqrt{N} \rceil$. Applying (33), (19), and (41) with $\rho = (6s)^{-1}$ yields

$$B_k^+(N) \leq (6s)^k P_k \left(\frac{5s}{36} + \frac{1}{5} \right) \qquad (2 \leq k \leq s). \tag{42}$$

For the tempered branch, use

$$\rho_B = \frac{17}{45k} = \frac{\beta}{6k}, \qquad \rho_Q = \frac{34}{45j} = \frac{\beta}{3j}.$$

The gas estimate (40) gives

$$B_k^+(N) \leq \left(\frac{45k}{17} \right)^k P_k \left(\frac{17N}{54k} + \frac{4N}{25k} \right), \tag{43}$$

$$Q_j(N) \leq \left(\frac{45j}{34} \right)^j P_j \left(\frac{17N}{108j} + \frac{2N}{25j} \right). \tag{44}$$

The normalization uses the lower bound in (19) and the factorial inequality

$$\left(\frac{25r}{68}\right)^r \leq r!, \tag{45}$$

proved by induction on r from the rational bound $(1+1/r)^r \leq 68/25$, a rational replacement for $(1+1/r)^r < \mathrm{e}$. The choice (37) is adapted to this estimate because

$$\frac{15}{34}\frac{68}{25} = \frac{6}{5}.$$

Consequently the residual powers are absorbed by (35).

Theorem 6.2 (Direct product bounds)**.** *Let $j = a - k$ (so $k + j = a$) and $N \leq 12a$, and recall $Q_j(N) \geq 0$.*

If $s \geq 32$, $2 \leq k \leq s$, $N \leq s^2$, and $j \geq 40$, then

$$\begin{aligned} 2\,2^j B_k(N)Q_j(N) \leq{}& \frac{2592}{25} kjc_kc_j \\ &\times P_{2a}\left(\frac{41}{36}s + \frac{j}{5} + \frac{29}{10}\frac{a}{j} + \frac{1}{5}\right). \end{aligned} \tag{46}$$

If $k, j \geq 40$, then

$$\begin{aligned} 2\,2^j B_k(N)Q_j(N) \leq{}& \frac{2592}{25} kjc_kc_j \\ &\times P_{2a}\left(\frac{a}{5} + \frac{57}{10}\frac{a}{k} + \frac{29}{10}\frac{a}{j}\right). \end{aligned} \tag{47}$$

Proof. For (46), combine (42) and (44), normalize by c_kc_j, absorb $s^k/k!$ and $(6/5)^j$ using (36) and (35), and apply Lemma 6.1 repeatedly. The Q-gas satisfies

$$\frac{17N}{108j} + \frac{2N}{25j} \leq \frac{29}{10}\frac{a}{j}$$

because $N \leq 12a$. The coefficient of s is $1 + 5/36 = 41/36$.

For (47), combine (43) and (44). Since $k + j = a$, both residual powers combine to $(6/5)^a$, which is absorbed into $P_a(a/5)$. The gas estimates

$$\frac{17N}{54k} + \frac{4N}{25k} \leq \frac{57}{10}\frac{a}{k}, \qquad \frac{17N}{108j} + \frac{2N}{25j} \leq \frac{29}{10}\frac{a}{j}$$

complete the proof. The product of the two normalization constants is $(36/5)(72/5) = 2592/25$. □

7 The positive budget for $a \geq 401$

We now insert Theorem 6.2 into the normalized majorant. Let

$$N_- = 6a - 7, \qquad N_+ = 12a - 8, \qquad s_- = \left\lceil \sqrt{N_-} \right\rceil, \qquad s_+ = \left\lceil \sqrt{N_+} \right\rceil, \qquad j = a - k.$$

For a rational $x < T$, define the certified exponential upper bound

$$\mathsf{E}_T(x) = \sum_{r=0}^{T-1} \frac{x^r}{r!} + \frac{x^T}{T!}\frac{1}{1 - x/T}. \tag{48}$$

The last term is a geometric majorant for the remaining exponential tail.

For $401 \le a \le 2000$, define

$$E_s(a,k) = \frac{1139}{1000}s_+ + \frac{j}{5} + \frac{29}{10}\frac{a}{j} + 1, \tag{49}$$

$$E_t(a,k) = \frac{a}{5} + \frac{57}{10}\frac{a}{k} + \frac{29}{10}\frac{a}{j} + 2. \tag{50}$$

For $401 \le a \le 2000$ and every retained k one has $0 \le E_s(a,k) < 800$. In the tempered regime $s_- < k$ — and hence whenever $k > \lceil \sqrt{N} \rceil$ for an admissible N, which is where E_t is used — one also has $0 \le E_t(a,k) < 800$. (Outside that regime E_t can be far larger, e.g. $E_t(2000,1) \approx 1.18 \cdot 10^4$, but it is never invoked there.) Thus each occurrence of E_{800} in (48) is well defined in the branch where it is used. The direct product theorem and elementary rational absorptions give

$$X_k(N)Y_j(N) \le \frac{2581}{20}\frac{kj}{NN_+}\mathsf{E}_{800}(E_s(a,k)), \qquad k \le \lceil \sqrt{N} \rceil, \tag{51}$$

$$X_k(N)Y_j(N) \le \frac{2117}{20}\frac{kj}{N^2}\mathsf{E}_{800}(E_t(a,k)), \qquad k > \lceil \sqrt{N} \rceil. \tag{52}$$

For the small branch, the comparison

$$\frac{2592}{25}N_+ \le \frac{5}{3}\frac{2581}{20}N$$

uses the lower edge $N \ge N_-$; the factor $5/3$ is absorbed as $P_1(2/3)$. The numerical coefficient $1139/1000$ exceeds $41/36$ by $1/9000$. For the tempered branch, $2592/25 < 2117/20$, and the extra $+2$ in (50) provides ample argument slack.

Combining (23), (51), and (52), the retained positive convolution is bounded by a sum of explicit terms

$$M_s(a,k) = \frac{65}{N_+}\frac{kj}{(a-1)\binom{a-2}{k-1}}2^{-j}\mathsf{E}_{800}(E_s(a,k)), \tag{53}$$

$$M_t(a,k) = \frac{96}{N_-}\frac{kj}{(a-1)\binom{a-2}{k-1}}2^{-j}\mathsf{E}_{800}(E_t(a,k)). \tag{54}$$

The branch union must respect the regime conditions: the small bound (51) applies only for $k \le \lceil \sqrt{N} \rceil$ and the tempered bound (52) only for $k > \lceil \sqrt{N} \rceil$, and $\lceil \sqrt{N} \rceil$ moves with N across $s_- \le \lceil \sqrt{N} \rceil \le s_+$. Outside its regime M_t has no content and can be large, so one may not simply take the unconditional maximum. The theorem-facing edge term is therefore the regime-aware

$$M_{\mathrm{edge}}(a,k) = \begin{cases} M_s(a,k), & k \le s_-, \\ \max\{M_s(a,k),\ M_t(a,k)\}, & s_- < k \le s_+, \\ M_t(a,k), & s_+ < k, \end{cases} \tag{55}$$

which dominates the retained summand for every admissible N: if $k \le \lceil \sqrt{N} \rceil$ then $k \le s_+$ and the small bound, maximized at the upper edge $N = N_+$, applies; if $k > \lceil \sqrt{N} \rceil$ then $s_- < k$ and the tempered bound, whose prefactor is maximized at the lower edge $N = N_-$, applies; only in the overlap $s_- < k \le s_+$ can the applicable branch depend on N, and there the maximum of the two is taken.

7.1 The finite strip $401 \le a \le 2000$

The formal certificate proves

$$\sum_{k \text{ retained}} M_{\text{edge}}(a,k) \le \frac{1}{200000000} \tag{56}$$

for every $401 \le a \le 2000$. The calculation is split into sixteen consecutive blocks of one hundred rows. Each block evaluates a verified upward-rounded fixed-point upper bound for (48) at scale 10^{20}. The executable Boolean is connected to the rational inequality (56) by a proved soundness theorem. Thus the displayed scan is not an external numerical assumption; it is the paper-level description of the Lean certificate interface. The prose reductions in Sections 2–7 are meant to explain the mathematics, while the Lean development is authoritative for the final constants and case splits.

The solo term in (14) is handled separately. The formal proof uses a sharpened factorial-composition estimate and proves

$$2^{-a-1} Y_a(N) \le \frac{1}{200000000} \tag{57}$$

throughout the same rectangle. This is the corrected theorem-facing bound; it does not use the obsolete shortcut $\exp(-0.49a)$ from an earlier draft. Thus the total positive envelope is at most

$$\frac{1}{200000000} + \frac{1}{200000000} = 10^{-8}. \tag{58}$$

7.2 The prefix strip $2001 \le a < 3000$

The same product inequalities are proved pointwise in this strip, with cutoffs a and $8a$ in the rational exponential-tail bounds. Generated prefix ratio and raw-budget certificates cover the finite lower-tempered transition that is inconvenient for a single asymptotic monotonicity argument. The solo estimate is supplied by the same sharp rational factorial-composition route used in the tail. No grid of exact $B_k(N)Q_j(N)$ products is expanded.

7.3 The analytic tail $a \ge 3000$

For the infinite range, the direct saddle bounds imply raw denominator-cleared versions of (51) and (52), with constants 130 and 192. The reciprocal binomial factor is controlled by

$$\binom{n}{k} \ge \left(\frac{n}{k}\right)^k, \tag{59}$$

and by a rational entropy-shadow inequality derived from the maximal weighted binomial term. The retained range is divided into a small branch and lower, middle, and upper tempered bands. Within each band, adjacent-term quotients are bounded by explicit rational numbers below one, and the first or last term of the band is multiplied by the corresponding geometric reserve $1/(1-\varrho)$. The concrete constants built into the formal tail are the following.

band	direction	adjacent-term ratio ϱ	boundary term initialized
small ($k \le \lceil \sqrt{N} \rceil$)	forward	$\le \frac{1}{2}$	first term ($k = 1$)
tempered, lower/middle	forward	$\le \dfrac{4a-1}{4a}$	first term of the band
tempered, upper	reverse	$\le \dfrac{4a-1}{4a}$	last term ($k = \lfloor 0.9a \rfloor$)

The tempered range is split at the natural-number threshold $k = \lfloor a/3 \rfloor + 10$, and each band's geometric reserve is allocated inside the half-budget $\frac{1}{2} \cdot 10^{-8}$. The ratios above are taken along the *candidate-majorant* sequence used to bound the band; the "first term ($k = 1$)" is the first term of that majorant sequence, whereas the actual guarded convolution term at $k = 1$ vanishes ($X_1 = -1$, so X_1 contributes nothing to the positive sum). The solo term is separately bounded through a $(10/7)^a$ envelope, as described below.

The lower-tempered prefix $2001 \le a < 3000$ is represented by generated quotient and raw-budget chunks; from $a = 3000$ onward, a symbolic sharp top-offset quotient theorem applies. The upper band is treated in the reverse direction. Separate unit-reserve theorems prove that the geometric multipliers fit inside the allocated half-budget. This hybrid construction proves the same uniform estimate as (56) for every $a > 2000$. The exact scalar inequalities behind this ratio–reserve argument — the candidate summands, the adjacent-ratio bounds, the endpoint reserves, and the $(10/7)^a$ solo envelope — are collected in Appendix A.

The solo term is bounded by splitting a closed factorial-composition sum into low-degree and remainder blocks, proving upper-edge monotonicity in N, and comparing the result to a $(10/7)^a$ envelope. The remaining dyadic factor then yields (57). All steps are rational inequalities; no floating-point assertion is imported into the theorem.

We summarize the outcome.

Theorem 7.1 (Positive envelope)**.** *For every $a \ge 401$ and every N satisfying* (5)*,*

$$2^{-a-1} Y_a(N) + \frac{N}{2} \sum_{\substack{1 \le k < a \\ X_k(N) > 0}} R_{k,a} 2^{-(a-k)} X_k(N) Y_{a-k}(N) \le 10^{-8}. \tag{60}$$

8 Completion of the proof

For $a \ge 401$, the rectangle gives $N \le 12a - 8 < (40/3)a$, so Theorem 5.1, applied with $m = a$, yields

$$X_a(N) \le -\left(\lambda\left(1 - \frac{2}{a}\right) - \frac{2215}{a^2}\right). \tag{61}$$

An exact endpoint comparison at $a = 401$, followed by monotonicity, proves

$$10^{-8} < \lambda\left(1 - \frac{2}{a}\right) - \frac{2215}{a^2}. \tag{62}$$

Combining (14), Theorem 7.1, and (62) gives $U_a(N) < 0$ for every $a \ge 401$.

Together with the direct partition enumeration for $a \le 8$, the exact majorant certificate for $9 \le a \le 60$, and the dyadic interval certificate for $61 \le a \le 400$, this proves Theorem 1.1.

9 The corrected Proposition 5.2

The companion residue class $g \equiv 1 \pmod 3$ is governed by Chen–Larson Proposition 5.2, which replaces the quotient B_μ by a derivative-corrected numerator. We record a correction to the printed relation, reduce the corrected coefficient to the Proposition 5.1 analysis by a one-line identity, and close the finitely many remaining degrees by an exhaustive check. Throughout, $g = 3a - 2$, $M := 2g - 2 = 6a - 6$, $q_i := m_i + 1 \ge 2$, $N := \sum_i q_i = M + n$, $s_j := \sum_i q_i^{-j}$, and $B_\mu(t) = \prod_i \mathsf{C}(t/q_i)/\mathsf{C}(t)^N$, $b_a(\mu) = [t^a]B_\mu(t)$ denote the Proposition 5.1 quotient series and its coefficient.

9.1 The correction

Ionel's relation [2], specialized to the degree $b = 1$ relevant here, contributes the factor

$$\kappa_0 - 2\kappa_1 t - 12 \sum_{j \geq 1} j c_j \kappa_{j+1} t^{j+1}, \tag{63}$$

where on $\mathcal{M}_g$ one has $\kappa_0 = 2g - 2 = M$. The series printed in Chen–Larson Proposition 5.2 instead has constant term 1. Dividing the whole relation by $\kappa_0 = M$ would be legitimate, but it would divide *every* term of (63); replacing only κ_0 by 1 is therefore not a normalization, and it changes the coefficient under consideration. Writing

$$\Phi(t) = 2t + 12t^2 \frac{\mathsf{C}'(t)}{\mathsf{C}(t)},$$

and using the stratum pullback $\iota^* f^* \kappa_j = (N - s_j)\eta^j$ for $j \geq 1$, the factor (63) pulls back to $M - K_\mu(\eta t)$ with

$$K_\mu(t) = \sum_i (q_i - 1)\, \Phi(t/q_i) = \sum_i m_i\, \Phi(t/q_i), \tag{64}$$

while the exponential prefactor pulls back to $B_\mu(\eta t)$. The corrected geometric reduction is therefore

$$\eta^a\, [t^a]\, B_\mu(t)\big(M - K_\mu(t)\big) = 0, \tag{65}$$

so $\eta^a = 0$ as soon as the scalar coefficient is nonzero. The scalar class $\kappa_0 = 2g - 2$ pulls back to M, not to 1; this is the entire content of the correction.

9.2 The decisive identity

Set

$$T_a^{\mathrm{old}}(\mu) := [t^a]\, B_\mu(t)\big(1 - K_\mu(t)\big), \qquad T_a^{\mathrm{cor}}(\mu) := [t^a]\, B_\mu(t)\big(M - K_\mu(t)\big)$$

for the printed and corrected coefficients. Subtracting and using $b_a(\mu) = [t^a] B_\mu(t)$ gives the exact identity

$$\boxed{T_a^{\mathrm{cor}}(\mu) = T_a^{\mathrm{old}}(\mu) + (M - 1)\, b_a(\mu).} \tag{66}$$

This is the point of the correction: it isolates the change as a multiple of the Proposition 5.1 coefficient $b_a(\mu)$, whose sign is already understood.

Remark 9.1 (Marked versus derivative form). The factor $M - K_\mu$ is the marked-numerator form of the geometric factor $M - 2A_\mu t - 12t^2 \mathcal{B}_\mu(t)$ read off from (63), where $A_\mu = N - s_1$ and $\mathcal{B}_\mu(t) = N \frac{\mathsf{C}'(t)}{\mathsf{C}(t)} - \sum_i \frac{\mathsf{C}'(t/q_i)}{q_i^2 \mathsf{C}(t/q_i)}$. The two forms give the same coefficient $[t^a] B_\mu(\cdot)$ because their difference contributes a multiple of $[t^{a-1}] B_\mu(t)$ that vanishes precisely when $M = 6a - 6$ — which is where the genus constraint $g = 3a - 2$ enters. The Lean development uses the marked form $M - K_\mu$ throughout.

9.3 Strict negativity for $a \geq 14$

Two sign inputs feed (66) in the uniform range.

The quotient coefficient. For a positive partition of $M = 6a - 6$ one has $N = M + n$ with $1 \le n \le M$, hence

$$6a - 5 \le N \le 12a - 12. \tag{67}$$

The Proposition 5.1 analysis of Sections 2–8 in fact proves a stronger *rectangular* statement: $b_a(\mu) < 0$ whenever $a \ge 9$ and $6a - 7 \le N \le 12a - 8$. Indeed, from the point at which the majorant $b_a(\mu) \le U_a(N)$ is introduced, no hypothesis on $\sum_i (q_i - 1)$ is used, and $U_a(N) < 0$ is established across that whole rectangle — exactly for $9 \le a \le 60$, by verified dyadic intervals for $61 \le a \le 400$, and analytically for $a \ge 401$. The interval (67) is contained in the rectangle, so

$$b_a(\mu) < 0 \qquad (a \ge 14). \tag{68}$$

(Formally this rectangular form is the theorem `Prop51.bCoeff_neg_of_rectangle`, obtained from the same certificates that prove Theorem 1.1.)

The printed coefficient. The printed-series coefficient is itself strictly negative in the uniform range,

$$T_a^{\mathrm{old}}(\mu) < 0 \qquad (a \ge 14), \tag{69}$$

a formal power-series statement independent of the geometric correction. Its proof splits at $a = 149$: an exact dyadic-interval certificate for $14 \le a \le 149$, and, for $a \ge 150$, a Gamma-integral and Taylor-truncation estimate of the normalized coefficient, reduced to a closed real exponential power-series identity. The whole argument is formalized in Lean — `Prop52.printedCoeffNegativityMid_closed` for the interval range and the `Prop52.Gamma`$\cdots$ modules for the tail; in particular the directed-rounding interval layer of the original Proposition 5.2 note is replaced by exact dyadic intervals checked by `native_decide`, so that this input too sits inside the formal trust boundary of Section 10.

Conclusion. For $a \ge 14$, combining (66), (68), (69), and $M - 1 > 0$ gives

$$T_a^{\mathrm{cor}}(\mu) = T_a^{\mathrm{old}}(\mu) + (M - 1)\, b_a(\mu) < 0.$$

The correction moves the coefficient further into the negative half-line: no new large-a cancellation has to be controlled.

9.4 The finite range $2 \le a \le 13$ and the theorem

It remains to prove $T_a^{\mathrm{cor}}(\mu) \ne 0$ for $2 \le a \le 13$, where the two summands of (66) need not share a sign. The coefficient is a rational number assembled from the finite recurrences for c_r, the exponential coefficients b_r, and the marked coefficients k_r, via $T_a^{\mathrm{cor}} = M b_a - \sum_{j=1}^{a} k_j b_{a-j}$. For $2 \le a \le 8$ the Lean proof evaluates the exact rational coefficient over every partition by `native_decide`; for $9 \le a \le 13$, where the partition counts run into the millions, it reduces each coefficient modulo a large prime and certifies a nonzero residue — a rational with a nonzero reduction at a prime not dividing its denominator cannot vanish — with the reduction map to $\mathbb{Q}$ proved in Lean.

Theorem 9.2 (Corrected Proposition 5.2, non-vanishing)**.** *For every integer $a \ge 2$ and every positive partition μ of $M = 6a - 6$, the corrected coefficient $T_a^{\mathrm{cor}}(\mu)$ is nonzero; for $a \ge 14$ it is strictly negative. This is Theorem 1.3.*

Proof. For $a \ge 14$ this is the conclusion of Section 9.3; for $2 \le a \le 13$ it is the finite certificate above. $\square$

Remark 9.3 (The correction is not cosmetic). For $g = 4$ ($a = 2$, $M = 6$) and $\mu = (1^6)$ one finds $b_2 = -195/8$, $T_2^{\mathrm{old}} = 45/8$, and $T_2^{\mathrm{cor}} = -465/4$, consistent with (66) since $-465/4 - 45/8 = 5 \cdot (-195/8) = (M-1)b_2$. The printed and corrected coefficients are not scalar multiples — here they even have opposite signs — so the correction is not a harmless renormalization. These three values are checked by `native_decide` in `Prop52/Theorem.lean`.

Remark 9.4 (Scope). The non-vanishing theorem is proved for positive integral partitions $m_i \geq 1$. The corrected geometric pullback can be written in a broader rational-part setting, but no global sign statement is claimed there.

10 Formalization architecture and trust

10.1 Public statement layer

The repository now has a concise public facade. Its mathematical content is:

1. the definition of the hypergeometric power series C;
2. the coefficient identity for C;
3. the quotient-series identity
$$\mathsf{C}(t)^N B_\mu(t) = \prod_i \mathsf{C}(t/(m_i+1));$$
4. strict negativity of the coefficient in Theorem 1.1;
5. the non-vanishing corollary.

The bridge from the recurrence-defined c_r to the formal power series is proved by showing that both C and $\exp(\sum_{r\geq 1} c_r t^r)$ solve the same formal differential equation with constant term one.

For the corrected Proposition 5.2 the public facade is `Prop52/Theorem.lean`. It defines the corrected coefficient $T_a^{\mathrm{cor}} = Mb_a - [t^a]B_\mu K_\mu$ (as `correctedCoeff`), records the decisive identity (66) (`correctedCoeff_eq_printedCoeff_add`), and exposes the two assumption-free theorems `correctedCoeff_nonvanishing` (every $a \geq 2$) and `correctedCoeff_neg` ($a \geq 14$), together with the $g = 4$ executable checks. It imports `Prop51.bCoeff_neg_of_rectangle` for the quotient sign and the `Prop52` proof library — the finite and modular certificates and the printed-coefficient interval and Gamma-tail arguments — for the analytic inputs. Both public theorems carry the same axiom report as the Proposition 5.1 facade.

10.2 Proof layers

The production proof has the following conceptual dependency chain:

layer	role
series bridge	identifies the recurrence-defined coefficients with the Chen–Larson generating series
majorant	replaces the partition-dependent numerator by $Q_r(N)$ and proves $b_a(\mu) \le U_a(N)$
finite certificates	handles $a \le 400$ by enumeration, exact rationals, and verified dyadic intervals
sign lock	eliminates $k > \lfloor 0.9a \rfloor$ and supplies the negative margin
direct saddle	proves the finite-prefix product inequalities (46) and (47)
positive assembly	checks the bounded edge budget and closes the prefix and infinite tail by ratio/reserve estimates
completion	combines all ranges into the unconditional public theorem

10.3 Trust boundary

Almost all of the argument is checked by Lean's small logical kernel, the same trusted core that underlies every Mathlib proof. The exception is the family of large finite computations — the enumerations and the exact and interval certificates — which are run by `native_decide`: Lean compiles the finite Boolean check to machine code, executes it, and accepts the result. This is faster by orders of magnitude than kernel reduction and is what makes the millions of certificate cases feasible, but it widens the trusted base to include the compiler for those steps. Following the Lean reference manual [3], each such use is recorded by a named axiom, and for the present development the complete list of extra-logical assumptions is exactly two: `Lean.ofReduceBool` and `Lean.trustCompiler`. Nothing else — in particular no `sorry`, no project-specific axiom, and no floating-point assumption — enters the final theorem.

This is verifiable in one command. The repository ships a short audit script that prints, for the series bridge, the majorant inequality, the Proposition 5.1 negativity theorem and non-vanishing corollary, and the corrected Proposition 5.2 theorems, the full list of axioms each depends on; a reader can reproduce it from a clean checkout with

```
lake exe cache get   # download the prebuilt Mathlib
lake build           # check every proof, including the certificates
lake env lean scripts/PublicAxiomsReport.lean   # print the axiom lists
```

The completed proof and the compact public facade are preserved as tagged releases in the repository. As an additional audit, the external checker `lean4checker` can replay the ordinary (non-native) proof objects and can detect certain bugs involving import state or declaration processing. It does not, however, remove the compiler trust attached to the native-evaluation certificates: proofs that depend on `Lean.trustCompiler` cannot at present be validated by an external proof checker without first replacing those computations by kernel-checkable certificates [3].

11 Background and use of generative artificial intelligence

This project was carried out almost entirely through generative AI systems, and the way they were used is part of what the paper reports. It began after Hannah Larson's talk *Tautological classes on the strata of differentials* in the Algebraic Geometry and Moduli seminar at ETH Zurich on 10 June 2026, which described the conjecture of Chen and Larson. The author was in the audience and decided to attack the resulting power-series problem with general-purpose AI tools. The human role throughout was orchestration rather than mathematics: choosing the problem, designing and relaying prompts, running and maintaining the computational and Lean environment, deciding from high-level signals when an autonomous run had stalled and an outside review should be solicited, and taking final responsibility for this text. The author did not originate or modify a mathematical lemma, estimate, proof strategy, or Lean proof. The proof search, much of the formal development, and parts of this write-up were produced with substantial AI assistance; no theorem-level step in the final argument was accepted until it had been formalized in Lean or independently checked as mathematics.

We document this as a chronological sequence of *human–AI interaction cards*, following the format proposed by Feng et al. [5]. Each card states the task, the model, the essential prompt, a one-line summary of the reply, and the human action it led to; the card title links to the public conversation where one exists. The prompts are reproduced verbatim or in close paraphrase, because the prompting strategy is the part of the method most likely to transfer to other problems; the AI replies are only summarized, with the most relevant exchanges also archived alongside the source.

In the terminology of Feng et al. [5] we classify the project as **Level A**, essentially autonomous, in its *content-based* sense: no human supplied or modified a mathematical lemma, estimate, proof strategy, or Lean construction. Human orchestration was nevertheless important to the workflow. The author selected the problem, designed and relayed prompts, maintained the computational environment, chose when to solicit independent AI reviews on the basis of high-level progress signals (for instance, an autonomous run that had continued for two days without converging), and took final editorial responsibility. In particular the two interventions recorded in Cards 6 and 7 were each a diagnosis produced by a *separate* AI model, which the author relayed to the worker without examining the mathematics. Thus "essentially autonomous" here refers to the generation of the core mathematical content, not to unattended end-to-end execution. As to significance, the theorem settles the coefficient nonvanishing that Chen and Larson isolate as their Proposition 5.1; we provisionally place the result at the boundary between significance Levels 1 and 2 of [5], with final classification left to expert peer review. In every case it is the Lean proof, not its provenance, that certifies the theorem.

The development had a simple shape, worth stating before the details. A single ChatGPT 5.5 Pro conversation (Card 1) was the central strand in which the mathematics was worked out and successively written up as LaTeX and Python. Into that strand the author repeatedly fed two distinct kinds of input from separate conversations — adversarial *audits* to surface errors (Card 2) and strategic *guidance* from a second model to suggest pivots (Card 3) — interleaved with simple requests to continue, producing ten drafts before formalization began. The formalization then moved into Lean, first broadly (Card 4) and then under a long-horizon autonomous goal (Card 5), with two moments (Cards 6 and 7) at which the author, prompted by the apparent stalling of the formalization attempt, solicited an outside AI review and relayed its diagnosis to redirect the search.

Card 1: Central investigation strand **ChatGPT 5.5 Pro**

Task. Attack the Chen–Larson coefficient problem from scratch and develop a candidate proof.
Prompt.

> *Can you have a look at Conjecture 1.4 of* arXiv:2603.23850 *(and its explicit reformulation Proposition 5.1) and make a serious, sustained, high-effort research attack at trying to crack it? This is an open problem, but in similar conversations you already managed to crack several such problems, and you might have a perspective on this power series that is not available easily to the authors. Would be really curious if you can make progress here!*

Reply (summary). Proposed the coefficientwise majorization that removes the partition, the normalized coefficients X_r, Y_r, and the strategy of a negative leading term against small positive corrections.
Continuation. This single conversation became the central strand: in later turns it produced the LaTeX write-up and Python verification scripts, and carried the mathematics through ten successive drafts, fed by the audits and guidance of Cards 2 and 3.

Card 2: Adversarial mathematical audit **ChatGPT 5.5 Pro**

Task. Surface every defect in a draft package (notes and code) before trusting it.
Prompt.

> *Consider the appended zip file. Please read the contained mathematical notes in detail and inspect all relevant code files, then proceed with a full mathematical audit, going paragraph by paragraph to identify any mathematical errors, gaps in arguments, hand-waving, misapplications of known results, or mis-cited theorems — anything that could affect the validity of the presented arguments. Then give me a full report.*

Reply (summary). Returned concrete defects — a far-tail constant valid only in a smaller range, a positive-part bound vacuous at $a = 401$, a single-edge scan that missed the worst value of N — each then repaired.
Human action. Fed each reported defect back into the central strand for repair, and re-ran the audit on the next draft.
Technique. A fresh conversation used purely adversarially: an exhaustive, paragraph-level audit of notes and *code, repeated each revision.*

Card 3: Strategic guidance **Claude Fable 5**

Task. Get higher-level direction, not line edits, when a draft stalled.
Prompt.

> *I wanted to ask for advice about the appended pure mathematics research note. For the theorem it tries to prove it shows some partial progress, but gaps remain. Please look through it carefully, think hard, do your own explorations or search for sources, and then give me a report focused on the most helpful advice possible for a second AI which will be tasked to continue pushing for a solution — concrete pointers to results, obstructions, a higher-level pivot, anything that could help crack the problem.*

Reply (summary). Returned strategic guidance — which effective-constant program to pursue, the worst-case regimes to target, and higher-level pivots — phrased as instructions the authoring model could act on.
Human action. Forwarded the guidance into the central strand to shape the next revision.
Technique. A second model used not to find errors but to suggest direction, with the prompt explicitly asking for advice aimed at the next AI in the loop.

Card 4: First formalization push **Claude Fable 5 (Claude Code)**

Task. Begin the Lean development: the statement layer, the series bridge, the partition-free majorant, the finite certificates, and the sign lock.

Prompt (excerpt).

> *Following your advice, the author put a tenth revision package together... this could be the basis of a first push towards a formalization by you. Take the time to review any changes, then if you think the basis is solid enough please go ahead and start developing the Lean code here.... The goal would be to isolate the desired end theorem in a file with as few imports, notation etc. as possible, with a clear certificate that the proof is* sorry*- and axiom-free.*

Reply (summary). Produced the bridge identifying C *with* $\exp(\sum c_r t^r)$*, the majorant inequality, the exact and dyadic-interval finite certificates, and the rational toolkit underlying the sign lock.*

Human action. Reviewed and integrated the layers, then handed the remaining analytic tail to a dedicated completion run.

Card 5: Long-horizon completion drive **OpenAI Codex CLI**

Task. Drive the formalization to completion with minimal supervision.

Prompt.

> */goal Please continue working on the Lean formalization of the argument, until it is either finished or until you hit a serious obstacle which requires either mathematical input or technical assistance beyond the standard difficulties and problem solving of the formalization process.*

Reply (summary). Over many cycles of inspection, coding, compilation, benchmarking, and repair, the worker proposed proof decompositions, searched Mathlib, refactored, generated finite-certificate checkers, and profiled slow computations.

Human action. Monitored progress, spot-checked the Lean state, and intervened only at the obstacle the goal itself anticipated (Card 6).

Technique. A persistent goal that doubles as a completion condition *attached to the thread [6]; before this, continuation had to be re-prompted every twenty to thirty minutes.*

Card 6: "Is the worker stuck?" — a review of the process **ChatGPT 5.5 Pro; Claude Opus 4.8**

Task. Decide whether the multi-day Codex run was healthy exploration or a dead end, and how to finish.

Prompt (excerpt).

> *Please review both the proof and the formalization files in detail. (i) Do you see any mathematical errors or obstructions in the strategy, and are they fixable? (ii) The formalization has run with a Codex worker for about two days; is this the expected process of investigation and optimization, or is the worker stuck in a local minimum or dead end? (iii) Any advice on how to finish, which could be forwarded to the worker.*

Reply (summary). Run in parallel in both systems; the reviews found no fatal obstruction in the main strategy but flagged unresolved analytic estimates and a proliferation of proof interfaces, with the worker reshaping one product-tail reduction rather than proving it, and advised freezing the completion route around a single direct combined-product inequality.

Human action. Forwarded the advice to the worker and gathered the worker's own specific blocking questions for the deeper review of Card 7.

Technique. Reviewing the process*, not only the mathematics, and running the same review in two independent systems.*

Card 7: The correction that unstuck the proof **ChatGPT 5.5 Pro; Claude Opus 4.8**

Task. Resolve the worker's blocking questions about the large-tail product estimate.
Prompt (relayed, excerpt).

> *[Relaying the worker's own questions:] What is the most direct analytic proof of the sharp large-tail product target, preferably by modifying the existing split rather than introducing a large generated table?*

Reply (summary). The forwarded answer showed that the attempted route — bounding the two factors separately and multiplying — is not merely expensive but false *at the first large-tail cell* $(a, k) = (3000, 4)$*, off by about* $3.26 \cdot 10^{557}$ *because the split discards a factor* 2^{a-k}*; it prescribed the fix used in Section 6: keep the product together, bound it with finite exponential prefixes, prove the prefix-convolution inequality, and normalize only at the last step.*
Human action. Relayed the corrected route to the worker, which implemented it as the direct-saddle method; the proof closed shortly afterward.
Technique. Escalating the worker's own *questions to a fresh, stronger review, and letting a provably false target redirect the architecture.*

What the two interventions show. The episode in Cards 6 and 7 is worth keeping in the record. The autonomous worker had made genuine local progress around an interface that was globally false: separating the two factors and multiplying them loses the dyadic cancellation, and no amount of finite refinement could have repaired it. What exposed this was not a sharper estimate but the formal target itself, which refused to close. Formal verification thus did more than confirm a known proof: it prevented a plausible but invalid analytic architecture from being mistaken for a theorem, and it marked exactly the moment at which the autonomous worker needed an outside perspective. That perspective, too, came from an AI model; the author's part was the orchestration — reading a high-level signal that the run had stalled and routing the question to a fresh review — not any judgement about the mathematics. The mathematical content was AI-generated within a human-orchestrated multi-agent process, and the human decisions that mattered were about *when* the worker was stuck, not how to fix it.

Scope of this record. The seven cards are a curated, not exhaustive, account: they record the interactions that materially shaped the proof and omit many routine continuation prompts and unproductive sessions.

The write-up itself was likewise AI-produced, in three stages. The original technical research note — the tenth revision, focused purely on the mathematics — was authored by ChatGPT 5.5 Pro in the initial research strand of Card 1. After the formalization was complete, a fresh expository draft of the present paper was produced by ChatGPT 5.5 Pro in a new conversation, following the author's high-level instructions on organization and intended audience. The subsequent revisions — including those prompted by an external review of the release candidate — were carried out in Claude Code under the author's direction, together with further ChatGPT 5.5 Pro review. As with the mathematics, the author's contribution to the write-up was direction and editorial judgement rather than composition; and acceptance of each mathematical claim ultimately rested on its Lean formalization, not on any model's assurance.

11.1 Addendum: the corrected Proposition 5.2

The same methodology produced the companion result, with two features worth recording: an AI audit caught a genuine error in the *published* paper, and the formalization then ran to completion with no human intervention at all.

After the Proposition 5.1 result was shared with the authors — who were open to the approach and asked whether the remaining case $g \equiv 1 \pmod 3$ (Proposition 5.2) could now be attacked — the author posed that problem to a ChatGPT 5.5 Pro conversation,[2] which produced a candidate non-vanishing proof for the *printed* Proposition 5.2 within three rounds. A separate ChatGPT 5.5 Pro audit of that candidate[3] flagged that the derivation of the series in Proposition 5.2 was itself in error: the constant $\kappa_0 = 2g - 2$ of Ionel's relation had been replaced by 1. An independent AI audit of the Chen–Larson paper alone[4] confirmed the error and supplied the corrected derivation of Section 9.1; asked to validate the point, the authors responded promptly that it was correct.

The corrected target was fed back into the original conversation, which — in two further rounds with intermediate feedback[5] — assembled a complete bundle: an informal proof of the corrected statement, an independent audit, the pinned Proposition 5.1 dependency, and a detailed Lean formalization plan with milestones. That bundle was handed to a fresh OpenAI Codex CLI worker, which completed the entire Lean formalization — the rectangle export, the correction identity, the finite and modular certificates, the in-Lean re-proof of the printed-coefficient sign, and the two assumption-free public theorems — *autonomously, with no further mathematical or technical intervention*, in roughly twenty hours. In the classification of [5] this is again Level A, and the human role was smaller still than for Proposition 5.1, reducing to relaying the problem, the audit verdict, and the finished bundle.

Two points deserve emphasis. First, the error was in the *published mathematics*, not in any formalization: the series printed in Proposition 5.2 is a valid power series, but it is not the one Ionel's relation produces, so a formal proof about the printed coefficient would have certified a true statement that does not carry the intended geometric meaning. The defect was surfaced by adversarial AI review of the paper itself — exactly the kind of error formal verification does not catch, because verification certifies *what one states*, while deciding what to state still needs independent mathematical judgement. Second, the contrast with Proposition 5.1 is instructive: given a sufficiently detailed, AI-authored formalization plan, the autonomous worker needed none of the mid-run redirection (Cards 6 and 7) that the first development had required.

Quantitative notes. The "roughly sixty-four hours" attributed to the completion phase is the elapsed wall-clock time recorded in the Codex goal log for that thread; it is not active compute time, is not summed over parallel workers, and includes waiting and compilation. Human active time was not separately measured, and token counts and monetary cost were not retained. The figure is given only to convey scale. The formal proof was checked under Lean 4.27.0 and a Mathlib revision pinned in the package manifest.

On the persistent goal. The `/goal` prompt of Card 5 is reproduced as it was actually used. By later best-practice standards [6] it is deliberately minimal: it states the desired outcome and the condition under which to stop and ask for help, but not the verification command, the permitted files, or an iteration policy. In practice the evolving Lean goal state, successful compilation, and the axiom reports served as the implicit evidence surface; a reproduction attempt should make those conditions explicit.

[2] `https://chatgpt.com/share/6a3bc8c9-e06c-83eb-b3e5-d098bb04573f`
[3] `https://chatgpt.com/share/6a3bc91c-0460-83eb-857d-d821168e38b7`
[4] `https://chatgpt.com/share/6a393ddd-bbc0-83eb-aa30-44806c9edc58`
[5] `https://chatgpt.com/share/6a3bc93b-3c68-83eb-8735-cb143fe10b12`

Limitations. Formal verification establishes the formal theorem relative to the trust boundary of Section 10. It does not by itself establish novelty or significance, the correctness of bibliographic claims, or the faithfulness of every sentence of the informal paper to the formal definitions; those remain matters of human and community judgement. The development also illustrates characteristic failure modes of AI-assisted research, which this record deliberately preserves: plausible but false intermediate estimates, a proliferation of proof interfaces, and convincing local progress on a globally false target. Curated, redacted records of the command-line-agent sessions — every on-topic human instruction from the selected sessions, with the AI side summarized — together with the most relevant review exchanges are archived under `docs/ai-correspondence/`, and the card titles link to the primary public conversations, so that the provenance can be inspected directly.

12 Reproducibility and repository organization

The repository exposes a small public API:

`Prop51/Theorem.lean`, `Prop52/Theorem.lean`, `Prop51.lean`, `Prop52.lean`.

The file `Prop51/Theorem.lean` is intended as the first audit surface: it names the series `C`, states the quotient-series identity, and exposes the strict-negativity and non-vanishing theorems for Proposition 5.1. Its counterpart `Prop52/Theorem.lean` exposes the corrected Proposition 5.2 coefficient and its two assumption-free theorems. The root files `Prop51.lean` and `Prop52.lean` import the respective libraries.

The proof backend remains more historical. In the current release, `Completion.lean` imports `OpenGoals.lean`, which contains both the live direct-saddle assembly and older certificate interfaces retained for audit and development history. This does not affect the public theorem or its axiom report, but a future presentation-only refactor can split the live direct-saddle construction into a small `Assembly.lean` and move the historical interfaces to a legacy namespace. Tagged releases in the repository preserve both the completed proof state and the public-facade state.

The release records the Lean toolchain, Mathlib revision, generated certificate provenance, and build commands. Continuous integration is set to run on both `main` and `master`; the default `lake build` now covers both the Proposition 5.1 and Proposition 5.2 libraries; the compact public axiom report (covering both facades) is run in CI; and internal AI correspondence is kept in a development archive rather than at repository root.

A Exact ratio and reserve estimates for the large positive tail

This appendix supplies the scalar inequalities behind the ratio–reserve argument of the large positive tail. Its purpose is twofold: to make the infinite summation in Section 7 reproducible without reading Lean code, and to record exactly where the finite prefix certificates enter. All quantities below are rational. We use the convention $0^0 = 1$.

A.1 Candidate summands

Fix $a > 2000$. Put

$$N_- = 6a - 7, \qquad N_+ = 12a - 8, \qquad s_- = \left\lceil \sqrt{N_-} \right\rceil, \qquad s_+ = \left\lceil \sqrt{N_+} \right\rceil, \tag{70}$$

and

$$K = \left\lfloor \frac{9a}{10} \right\rfloor, \qquad \ell = \max\{1, s_- + 1\}, \qquad m = \left\lfloor \frac{a}{3} \right\rfloor + 10, \qquad j_k = a - k. \tag{71}$$

For $a > 2000$ one has

$$1 \le \ell \le m < K < a. \tag{72}$$

The entropy-shadow factor is

$$\mathcal{H}_a(k) = \frac{(k-1)^{k-1}(j_k-1)^{j_k-1}}{(a-2)^{a-2}}. \tag{73}$$

The maximal-term estimate for the binomial expansion gives

$$\frac{1}{(a-1)\binom{a-2}{k-1}} \le \mathcal{H}_a(k) \qquad (1 \le k < a-1). \tag{74}$$

For $k = 1$ this is simply $1/(a-1) \le 1$. For $k \ge 2$, put $n = a-2$ and $r = k-1$. In the binomial expansion

$$n^n = \sum_{i=0}^{n} \binom{n}{i} r^i (n-r)^{n-i},$$

the summand at $i = r$ is maximal. Hence

$$n^n \le (n+1)\binom{n}{r} r^r (n-r)^{n-r},$$

which rearranges exactly to (74).

Recall the rational exponential upper bound

$$\mathsf{E}_T(x) = \sum_{q=0}^{T-1} \frac{x^q}{q!} + \frac{x^T}{T!}\frac{1}{1-x/T}, \qquad 0 \le x < T. \tag{75}$$

Write

$$E_s(a,k) = \frac{1139}{1000}s_+ + \frac{j_k}{5} + \frac{29}{10}\frac{a}{j_k} + 1, \tag{76}$$

$$E_t(a,k) = \frac{a}{5} + \frac{57}{10}\frac{a}{k} + \frac{29}{10}\frac{a}{j_k} + 2, \tag{77}$$

and set

$$L_s(a,k) = \mathsf{E}_a(E_s(a,k)), \qquad L_t(a,k) = \mathsf{E}_{8a}(E_t(a,k)). \tag{78}$$

The two candidate summands used in the formal tail proof are

$$S_a(k) = \frac{65}{N_+}\, k j_k\, \mathcal{H}_a(k)\, 2^{-j_k} L_s(a,k), \tag{79}$$

$$T_a(k) = \frac{96}{N_-}\, k j_k\, \mathcal{H}_a(k)\, 2^{-j_k} L_t(a,k). \tag{80}$$

They are not heuristic asymptotic expressions: they are the exact rational majorants summed by the Lean proof.

To relate them to the normalized convolution in (14), define

$$\mathcal{C}_a(N,k) = \frac{N}{2} R_{k,a} 2^{-j_k} \max\{X_k(N), 0\}\, Y_{j_k}(N). \tag{81}$$

The direct saddle estimates, the bound (23), and (74) imply

$$\mathcal{C}_a(N,k) \le S_a(k) \qquad \text{if } k \le \left\lceil \sqrt{N} \right\rceil, \tag{82}$$

$$\mathcal{C}_a(N,k) \le T_a(k) \qquad \text{if } \left\lceil \sqrt{N} \right\rceil < k. \tag{83}$$

The sign lock removes all $k > K$. Consequently, for every $N_- \le N \le N_+$,

$$\sum_{1 \le k < a} \mathcal{C}_a(N,k) \le \Sigma_s(a) + \Sigma_t(a), \tag{84}$$

where

$$\Sigma_s(a) = \sum_{k=1}^{\min\{K, s_+\}} S_a(k), \qquad \Sigma_t(a) = \sum_{k=\ell}^{K} T_a(k). \tag{85}$$

The overlap is deliberately counted twice. This loses a negligible amount and allows the two regimes to be summed independently.

A.2 The common adjacent quotient

Let $1 \le r < K$ and write $j = a - r$. Removing the constants and exponential shells from either (79) or (80) gives the same adjacent quotient

$$\mathfrak{q}_a(r) = 2\,\frac{(r+1)(j-1)r^r(j-2)^{j-2}}{rj(r-1)^{r-1}(j-1)^{j-1}}. \tag{86}$$

Thus

$$\frac{S_a(r+1)}{S_a(r)} = \mathfrak{q}_a(r)\frac{L_s(a,r+1)}{L_s(a,r)}, \qquad \frac{T_a(r+1)}{T_a(r)} = \mathfrak{q}_a(r)\frac{L_t(a,r+1)}{L_t(a,r)}. \tag{87}$$

For $r \ge 2$ (and hence, in the retained range, $j \ge 3$), it is useful to factor

$$\mathfrak{q}_a(r) = 2\frac{r+1}{j}\,\mathcal{P}_a(r), \tag{88}$$

where

$$\begin{aligned} \mathcal{P}_a(r) &:= \frac{r^r}{r(r-1)^{r-1}}\,\frac{(j-1)(j-2)^{j-2}}{(j-1)^{j-1}} \\ &= \left(1 + \frac{1}{r-1}\right)^{r-1}\left(1 + \frac{1}{j-2}\right)^{-(j-2)}. \end{aligned} \tag{89}$$

Since $n \mapsto (1 + 1/n)^n$ is increasing, one obtains

$$\mathcal{P}_a(r) \le 1 \quad \text{if } r-1 \le j-2, \qquad \mathcal{P}_a(r) \ge 1 \quad \text{if } j-2 \le r-1. \tag{90}$$

We repeatedly use the elementary geometric estimates

$$F_{r+1} \le qF_r \ (u \le r < v) \Longrightarrow \sum_{r=u}^{v} F_r \le \frac{F_u}{1-q}, \tag{91}$$

$$F_{r-1} \le qF_r \ (u < r \le v) \Longrightarrow \sum_{r=u}^{v} F_r \le \frac{F_v}{1-q}, \tag{92}$$

valid for nonnegative F_r and $0 \le q < 1$.

A.3 The small branch

Lemma A.1 (Small adjacent ratio)**.** *For $a > 2000$ and $1 \le r < \min\{K, s_+\}$, one has*

$$S_a(r+1) \le \frac{1}{2} S_a(r). \tag{93}$$

Proof. On this range the entropy-shadow quotient satisfies

$$\mathfrak{q}_a(r) \le \frac{1}{2}. \tag{94}$$

This is a denominator-cleared rational inequality obtained from (86) and the bounds $r < s_+$ and $s_+^2 \ge N_+ = 12a - 8$. The small exponent is nonincreasing:

$$E_s(a, r+1) \le E_s(a, r). \tag{95}$$

Monotonicity of (75) in its argument therefore gives $L_s(a, r+1) \le L_s(a, r)$. Substitution in (87) proves the claim. □

The first candidate has the exact form

$$S_a(1) = \frac{65(a-1)}{12a-8}\, 2^{-(a-1)} L_s(a, 1), \tag{96}$$

because $\mathcal{H}_a(1) = 1$. The exponential shell satisfies

$$L_s(a, 1) \le \left(\frac{3}{2}\right)^a. \tag{97}$$

Indeed $E_s(a, 1) \le 3a/10$, and the finite rational exponential bound at $3a/10$ is dominated by the negative-binomial envelope $(3/2)^a$. Therefore

$$800000000\, S_a(1) \le 800000000 \frac{65(a-1)}{12a-8} 2^{-(a-1)} \left(\frac{3}{2}\right)^a. \tag{98}$$

The right-hand side is nonincreasing for $a \ge 3$; its value at $a = 2001$ is at most one by an exact rational computation. Hence

$$800000000\, S_a(1) \le 1 \qquad (a > 2000). \tag{99}$$

Combining Lemma A.1, (91), and (99) gives

$$\Sigma_s(a) \le 2 S_a(1) \le \frac{1}{400000000}. \tag{100}$$

The actual guarded convolution term at $k = 1$ vanishes because $X_1 = -1$. The nonzero value $S_a(1)$ is retained only as a convenient first-term reserve for the candidate sequence.

A.4 The lower tempered band

Put

$$q_a = \frac{4a-1}{4a} = 1 - \frac{1}{4a}. \tag{101}$$

The lower tempered band is $\ell \le k \le m$. Its desired step estimate is

$$T_a(r+1) \le q_a T_a(r) \qquad (\ell \le r < m). \tag{102}$$

On this range $r-1 \le j-2$, so $\mathcal{P}_a(r) \le 1$ by (90), and therefore

$$\mathfrak{q}_a(r) \le 2\frac{r+1}{j}. \tag{103}$$

Define the sharp exponential quotient target

$$U_a^\sharp(r) = \frac{1}{2}\frac{j}{r+1}q_a. \tag{104}$$

Then (103) gives the exact budget identity

$$\mathfrak{q}_a(r)U_a^\sharp(r) \le q_a. \tag{105}$$

Thus it remains only to prove

$$\frac{L_t(a,r+1)}{L_t(a,r)} \le U_a^\sharp(r). \tag{106}$$

There are two subranges. If

$$3(r+1) \le a, \tag{107}$$

then

$$E_t(a,r+1) \le E_t(a,r), \qquad U_a^\sharp(r) \ge 1. \tag{108}$$

Hence (106) follows immediately from monotonicity of E_{8a}.

The only remaining indices are the ten offsets

$$r = \left\lfloor \frac{a}{3} \right\rfloor + t, \qquad 0 \le t < 10. \tag{109}$$

For $a \ge 3000$ the following two rational inequalities hold uniformly in t:

$$E_t(a,r+1) + \frac{175}{4a} \le E_t(a,r), \tag{110}$$

$$1 \le U_a^\sharp(r)\left(1 + \frac{175}{12a}\right)^3. \tag{111}$$

The second estimate is reduced to the nonnegativity of

$$550656a^3 - 76358800a^2 - 559763750a + 144703125. \tag{112}$$

After writing $a = 3000 + x$, this polynomial becomes

$$550656x^3 + 4879545200x^2 + 14408999436250x + 14178803653453125, \tag{113}$$

which is manifestly nonnegative for $x \ge 0$.

The finite-exponential three-step shift inequality says that, whenever $y + d \le z$,

$$\mathsf{E}_T(y) \le \left(1 + \frac{d}{3}\right)^{-3} \mathsf{E}_T(z) \tag{114}$$

provided the displayed arguments lie in the certified range. Applying it with $d = 175/(4a)$ and using (110)–(111) proves (106) for every $a \ge 3000$.

For the finite prefix $2001 \le a < 3000$, define

$$\theta_a = 1 - \frac{213}{5a} = \frac{5a - 213}{5a}. \tag{115}$$

Exact rational certificates verify, for every guarded pair (a, t) with (109),

$$\frac{L_t(a, r+1)}{L_t(a, r)} \le \theta_a, \tag{116}$$

$$(4a)\,\mathfrak{q}_a(r)\theta_a \le 4a - 1. \tag{117}$$

The first computation is split into twenty blocks of fifty consecutive values of a, each containing the ten offsets $t = 0, \ldots, 9$. The second is a single 999×10 raw-rational block. These are exact Boolean checks over $\mathbb{Q}$ executed by Lean's native evaluator; no floating-point comparison is used. Equations (116) and (117) imply (102) in the prefix strip. Together with the symbolic argument above, they prove (102) for every $a > 2000$.

A.5 The upper tempered band

The upper band is $m + 1 \le k \le K$ and is summed in reverse. The desired step is

$$T_a(r-1) \le q_a T_a(r) \qquad (m + 1 < r \le K). \tag{118}$$

Set

$$V_a^\sharp(r) = \mathfrak{q}_a(r-1) q_a. \tag{119}$$

By the exact quotient identity (87), it is enough to prove

$$L_t(a, r-1) \le V_a^\sharp(r) L_t(a, r). \tag{120}$$

In the far upper band

$$3a \le 5r, \tag{121}$$

one has

$$E_t(a, r-1) \le E_t(a, r), \qquad V_a^\sharp(r) \ge 1. \tag{122}$$

The second inequality follows from the lower form of (90), applied to the raw quotient at $r - 1$. Thus (120) is immediate in this range.

It remains to treat the middle band

$$m + 1 < r \le K, \qquad 5r < 3a. \tag{123}$$

The exponent shift is bounded by

$$E_t(a, r-1) \le E_t(a, r) + \delta_a, \qquad \delta_a = \frac{45}{a}. \tag{124}$$

Put

$$\Phi_a = \left(1 - \frac{79}{75}\delta_a\right)^{-1}. \tag{125}$$

The rational exponential estimate used in Lean is

$$\mathsf{E}_{8a}(E + \delta_a) - \mathsf{E}_{8a}(E) \le \frac{79}{75}\delta_a\, \mathsf{E}_{8a}(E + \delta_a), \tag{126}$$

valid here because

$$E + \delta_a \le \frac{8a - 1}{24}. \tag{127}$$

Rearranging (126) and using (124) gives

$$L_t(a, r - 1) \le \Phi_a L_t(a, r). \tag{128}$$

It remains to show $\Phi_a \le V_a^\sharp(r)$.

For this purpose write the raw quotient at $r - 1$ as

$$\mathfrak{q}_a(r - 1) = \frac{2r}{a - r + 1}\, \mathcal{P}_a(r - 1). \tag{129}$$

Define

$$W_a(r) = \Phi_a \frac{4a}{4a - 1}\frac{a - r + 1}{2r}. \tag{130}$$

Then $\Phi_a \le V_a^\sharp(r)$ is equivalent to

$$W_a(r) \le \mathcal{P}_a(r - 1). \tag{131}$$

There are two elementary subcases.

If $a + 1 \le 2r$, then direct rational simplification gives

$$W_a(r) \le 1, \tag{132}$$

and (90) gives $1 \le \mathcal{P}_a(r - 1)$.

If $2r < a + 1$, then

$$W_a(r) \le 1 - \frac{1}{a}. \tag{133}$$

The required matching lower bound for the power product is obtained from the telescoping inequality

$$\mathcal{P}_a(s) \ge 1 - \frac{(j_s - 2) - (s - 1)}{2(s - 1)(j_s - 2)} \qquad (s - 1 \le j_s - 2). \tag{134}$$

For $s = r - 1$ and the present range, the fraction on the right is at most $1/a$; hence

$$1 - \frac{1}{a} \le \mathcal{P}_a(r - 1). \tag{135}$$

This proves (131), then (120), and finally (118) throughout the upper band.

A.6 Endpoint reserves and the tempered budget

The two endpoint exponential shells admit the common envelope

$$L_t(a,\ell) \le \left(\frac{10}{7}\right)^a, \qquad L_t(a,K) \le \left(\frac{10}{7}\right)^a. \tag{136}$$

The theorem-facing endpoint estimates are

$$800000000\,(4a)T_a(\ell) \le 1, \qquad 800000000\,(4a)T_a(K) \le 1. \tag{137}$$

We record the coarse envelopes used to prove them.

For the lower endpoint, one has $\ell \le a/4$, discards $\mathcal{H}_a(\ell) \le 1$, and keeps only the dyadic decay. This yields

$$800000000(4a)T_a(\ell) \le G_-(a) := 3200000000\,a^3\,2^{-(a-\lfloor a/4\rfloor)}\left(\frac{10}{7}\right)^a. \tag{138}$$

The four-step comparison

$$G_-(a+4) \le G_-(a) \tag{139}$$

follows from

$$a+4 \le \frac{401}{400}a, \qquad 10000\left(\frac{401}{400}\right)^3 \le 19208. \tag{140}$$

Exact rational checks at $a = 2001, 2002, 2003, 2004$ initialize the four residue classes and prove the first inequality in (137).

For the upper endpoint, the entropy shadow is retained. The elementary bounds

$$\mathcal{H}_a(K) \le \left(\frac{901}{1000}\right)^{\lfloor 89a/100\rfloor}\left(\frac{101}{1000}\right)^{\lfloor a/10\rfloor-1} \tag{141}$$

give

$$\begin{aligned}&800000000(4a)T_a(K) \le G_+(a)\\ &\qquad := 3200000000\,a^3\left(\frac{901}{1000}\right)^{\lfloor 89a/100\rfloor}\left(\frac{101}{1000}\right)^{\lfloor a/10\rfloor-1} 2^{-\lfloor a/10\rfloor}\left(\frac{10}{7}\right)^a.\end{aligned} \tag{142}$$

The one-hundred-step comparison

$$G_+(a+100) \le G_+(a) \tag{143}$$

follows from

$$\left(\frac{21}{20}\right)^3\left(\frac{901}{1000}\right)^{89}\left(\frac{101}{1000}\right)^{10}\left(\frac{10}{7}\right)^{100} \le 2^{10}. \tag{144}$$

Exact rational checks for $2001 \le a \le 2100$ initialize the one hundred residue classes and prove the second inequality in (137).

Since $1-q_a = 1/(4a)$, the lower and upper geometric reserves satisfy

$$\sum_{k=\ell}^{m} T_a(k) \le \frac{T_a(\ell)}{1-q_a} = 4aT_a(\ell) \le \frac{1}{800000000}, \tag{145}$$

$$\sum_{k=m+1}^{K} T_a(k) \le \frac{T_a(K)}{1-q_a} = 4aT_a(K) \le \frac{1}{800000000}. \tag{146}$$

Therefore

$$\Sigma_t(a) \le \frac{1}{400000000}. \tag{147}$$

Together with (100), this gives the product part of the positive budget:

$$\Sigma_s(a) + \Sigma_t(a) \le \frac{1}{200000000} \qquad (a > 2000). \tag{148}$$

A.7 The solo term

The remaining term in (14) is $2^{-a-1}Y_a(N)$. The sharp solo estimate is proved first at the upper edge $N = N_+$ by splitting the closed factorial-composition expansion into low-degree and remainder blocks. In the notation of the formal development, the closed block sum is

$$\begin{aligned}\Delta_a(N) = &\sum_{s=0}^{a-1} \frac{(Nc_1/4)^s}{s!} \sum_{\substack{r\ge 1\\ 2r\le a-s}} \frac{(2N/25)^r\, 3^{a-s}\, 4^{r-1}\, (a-s-2r+1)!}{r!} \\ &+ \frac{(Nc_1/4)^a}{a!},\end{aligned} \tag{149}$$

which is `positiveLargeTailSoloSharpGcompClosedFactorialSplitBlockSum` in Lean. The denominator-cleared estimate at the upper edge is

$$4\,2^a \Delta_a(N_+) \le 29\, a\, c_a \left(\frac{10}{7}\right)^a. \tag{150}$$

Monotonicity in N then yields the normalized bound

$$Y_a(N) \le \frac{29}{2}\frac{a}{N}\left(\frac{10}{7}\right)^a \qquad (N_- \le N \le N_+). \tag{151}$$

Since $N \ge 6a - 7$ and $a \ge 401$, one has $a/N \le 1/5$. Therefore

$$\begin{aligned}2^{-a-1}Y_a(N) &\le \frac{29}{4}\frac{a}{N}\left(\frac{5}{7}\right)^a \\ &\le \frac{29}{20}\left(\frac{5}{7}\right)^a.\end{aligned} \tag{152}$$

Writing

$$H_{\mathrm{solo}}(a) = 290000000\left(\frac{5}{7}\right)^a,$$

one has the exact adjacent-a relation

$$H_{\mathrm{solo}}(a+1) = \frac{5}{7}H_{\mathrm{solo}}(a). \tag{153}$$

Thus the sequence is decreasing, and the exact rational check at $a = 401$ gives

$$290000000\left(\frac{5}{7}\right)^a \le 1. \tag{154}$$

Consequently

$$2^{-a-1}Y_a(N) \le \frac{1}{200000000}. \tag{155}$$

Proposition A.2 (Large positive-tail budget)**.** *For every $a > 2000$ and every N with $6a - 7 \leq N \leq 12a - 8$,*

$$2^{-a-1}Y_a(N) + \frac{N}{2} \sum_{\substack{1 \leq k < a \\ X_k(N) > 0}} R_{k,a} 2^{-(a-k)} X_k(N) Y_{a-k}(N) \leq 10^{-8}. \tag{156}$$

Proof. The sign lock removes $k > K$. The remaining product sum is at most (148) by (82)–(84). The solo term is at most (155). Their sum is $1/200000000 + 1/200000000 = 10^{-8}$. □

A.8 Correspondence with the formal proof

For audit purposes, the principal Lean declarations corresponding to this appendix are listed below. The declaration names are stable in the release version of the repository.

paper item	Lean declaration or file
L_s, L_t	`positiveSmallLargeExp`; `positiveTemperedLargeExp`
S_a, T_a	`positiveSmallEntropyShadowExpMajorantTerm`; `positiveTemperedEntropyShadowExpMajorantTerm`
raw quotient $\mathfrak{q}_a$	`positiveEntropyShadowBaseStepRawQuotient`
power product $\mathcal{P}_a$	`positiveEntropyShadowBaseStepRawPowerProduct`
small ratio 1/2	`positiveSaddleLargeTailCandidateSmallRawBaseHalfCertificate`
lower split $m = \lfloor a/3 \rfloor + 10$	`positiveLargeExpTemperedSplit`
lower sharp target $U_a^\sharp$	`positiveTemperedLowerSharpExpQuotientTarget`
finite prefix target θ_a	`positiveTemperedLowerPrefixTopOffsetExpRatioTarget`
finite prefix certificates	`PositiveSaddlePrefixExpRatioCertificate.lean`; `PositiveSaddlePrefixRawBudget.lean`
upper-middle shift $45/a$	`positiveTemperedUpperMiddleExpShiftBudget`
upper-middle factor Φ_a	`positiveTemperedUpperMiddleExpShiftFactor`
small first reserve	`positiveSaddleLargeTailCandidateSmallFirstReserveCertificate_closed`
tempered endpoint reserves	`positiveSaddleLargeTailCandidateUnitReserveCertificate_temperedTenSevenths_closed`
assembled hybrid ratio certificate	`positiveSaddleLargeTailCandidateRefinedAtomicCertificate_hybridClosed`
solo block sum Δ_a	`positiveLargeTailSoloSharpGcompClosedFactorialSplitBlockSum`
solo block-sum cleared	`positiveLargeTailSoloSharpGcompClosedFactorialSplitBlockSumTenSeventhsCleared`
solo saddle cleared	`positiveLargeTailSoloSharpGcompSaddleTenSeventhsCleared`
solo scalar budget	`positiveLargeTailSoloTenSeventhsScalarBudget_of_ge_401`

B Public Lean statement

The following is a Lean-style rendering of the public facade, lightly edited for readability (e.g. `Q[[X]]` for the rational power-series type and ASCII `<=`, `!=` for the order and disequality); it is not literal source. The exact, machine-checked file is `Prop51/Theorem.lean` in the repository. It is reproduced to make the correspondence between the paper and the formal theorem immediate.

```
import Prop51.BCoeffSeries
import Prop51.Completion

namespace Prop51

open PowerSeries

noncomputable abbrev chenLarsonC : Q[[X]] := Cseries

theorem coeff_chenLarsonC (k : Nat) :
    coeff k chenLarsonC =
      (Nat.factorial (6*k) : Q) /
        ((Nat.factorial (3*k) : Q) *
          (Nat.factorial (2*k) : Q) * 72^k)

noncomputable abbrev chenLarsonSeries (mu : List Nat) : Q[[X]] :=
  bSeries mu

theorem chenLarsonSeries_spec (mu : List Nat) :
    chenLarsonC ^ ((mu.map (fun m => m + 1)).sum) *
      chenLarsonSeries mu
      = (mu.map (fun m =>
          rescale (((m + 1 : Nat) : Q)^(-1)) chenLarsonC)).prod

theorem chenLarsonCoefficient_neg
    {g : Nat} (hg : 2 <= g) (hmod : g % 3 != 1)
    {mu : List Nat} (hmu : IsPartitionOf mu (2 * g - 2)) :
    coeff (g / 3 + 1) (chenLarsonSeries mu) < 0

theorem chenLarsonCoefficient_ne_zero
    {g : Nat} (hg : 2 <= g) (hmod : g % 3 != 1)
    {mu : List Nat} (hmu : IsPartitionOf mu (2 * g - 2)) :
    coeff (g / 3 + 1) (chenLarsonSeries mu) != 0

end Prop51
```

For the corrected Proposition 5.2, the analogous public facade is `Prop52/Theorem.lean`; the two final theorems read as follows (again lightly edited for readability).

```
import Prop52.Assembly

namespace Prop52

-- correctedCoeff a mu = M(a) * Prop51.bCoeff mu a - markedConvolution mu a
--                      = [t^a] B_mu(t) * (M - K_mu(t)),   with M = 6*a - 6
```

```
theorem correctedCoeff_nonvanishing :
    forall a, 2 <= a -> forall mu, Prop51.IsPartitionOf mu (M a) ->
      correctedCoeff a mu != 0

theorem correctedCoeff_neg
    {a : Nat} (ha : 14 <= a)
    {mu : List Nat} (hmu : Prop51.IsPartitionOf mu (M a)) :
    correctedCoeff a mu < 0

end Prop52
```

C Paper-to-Lean correspondence

paper item	principal Lean declaration or file
hypergeometric series C	`Aseq`, `Cseries` in `ExpSeries.lean`
$\mathsf{C} = \exp(\sum c_r X^r)$	`Cseries_eq_expSeries_c` in `Bridge.lean`
quotient-series identity	`bSeries_official` in `BCoeffSeries.lean`
partition-free majorant	`bCoeff_le_U` in `Majorant.lean`
coefficient bounds	`d_lb`, `d_ub`, `d_increment`, `d_ratio_lb` in `DNorm.lean`
sign lock	`Xnorm_le_neg_final_margin` in `SignLock.lean`
finite saddle and prefix convolution	`expCoeff_saddle`, `expPrefix_mul_le` in `SaddleDirect.lean`
direct product estimates	`Bq_Qq_small_core`, `Bq_Qq_tempered_core` in `SaddleDirect.lean`
finite edge budget	`positiveSaddleFiniteScaledEdgeBudget` in `Generated/PositiveSaddleFiniteScaledEdge.lean`
closed proof	`completion_of_directSaddle_closed` and `coefficientNegativity`
public theorem	`chenLarsonCoefficient_neg`, `chenLarsonCoefficient_ne_zero` in `Prop51/Theorem.lean`
Corrected Proposition 5.2	
corrected coefficient T_a^{cor}	`correctedCoeff` in `Prop52/Statement.lean`
central identity (66)	`correctedCoeff_eq_printedCoeff_add` in `Prop52/Assembly.lean`
rectangle quotient sign	`Prop51.bCoeff_neg_of_rectangle` in `Prop51/Rectangle.lean`
finite range $2 \le a \le 13$	`correctedCoeff_finite_nonvanishing` in `Prop52/Finite.lean`
printed sign, $14 \le a \le 149$	`printedCoeffNegativityMid_closed` in `Prop52/MidBridge.lean`
printed sign, Gamma tail	`printedTailERealSeriesHasSum` in `Prop52/GammaCompletion.lean`
public theorems (Prop. 5.2)	`correctedCoeff_nonvanishing`, `correctedCoeff_neg` in `Prop52/Theorem.lean`

D Computational certificate inventory

range	method	verification surface
$a \le 8$	exact partition enumeration	exact rational coefficient for every positive partition; selected partition-cardinality checks
$9 \le a \le 60$	exact rational majorant	all 10764 rectangle pairs
$61 \le a \le 400$	192-bit outward dyadic intervals	proved enclosure of recurrence tables; eight native chunks
$401 \le a \le 2000$	direct saddle plus fixed-point edge certificate	sixteen 100-row shards, scale 10^{20}; analytic solo theorem
$2001 \le a < 3000$	direct saddle plus generated prefix ratios	pointwise product and solo bounds; finite quotient/raw-budget chunks
$a \ge 3000$	symbolic rational tail	adjacent quotients, reverse upper band, unit reserves, and sharp solo envelope

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